Hard k-CNF instances for PPSZ with small resolution width and near-2/k savings upper bound

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Editorial scope. Single main mechanism: hard instances and bounded-width PPSZ success upper bound. Supporting graph/width/game lemmas, parameter consequences and common-lift proof grouped once. Source's own final density/padding detail omission preserved; no author proof repair, extra assumptions, unconditional ETH/SETH claim or new conjecture.

Allowed standard prerequisites (editorial).

- Large-girth regular graphs, Hall's theorem and the Atserias-Dalmau resolution-width game characterization. Formal Erdős–Sachs[4] regular-graph existence and Atserias–Dalmau Theorem9[2] remain cited standard prerequisites. The paper's common-lift proof is completely retained in AppendixA rather than merely cited. All12 original bibliography entries retained.

PPSZ and width-bounded proof heuristics (Introduction)

The PPSZ algorithm for k-SAT by Paturi, Pudlák, Saks, and Zane [7] is simple to state but famously difficult to analyze. Given a k-CNF formula Φ as input, it first chooses a random ordering π of its variables $x_1, \ldots, x_m$. It goes through them one by one, in the order given by π. For each variable x, it tries to derive the correct value using a certain proof heuristic P. P takes as input a k-CNF formula Φ and a variable x and returns a value in $\{0, 1, ?\}$. P must be sound, meaning if $P(\Phi, x) = b \in \{0, 1\}$ then $\Phi \models (x = b)$, i.e., every satisfying assignment of Φ sets x to b; however, we allow P to be incomplete, i.e., it may answer "?", meaning "I don't know". If $P(\Phi, x) = b \in \{0, 1\}$, then PPSZ sets x to b; otherwise it sets x to some $b \in \{0, 1\}$ chosen uniformly at random. In either case, it simplifies Φ to $\Phi|_{x \mapsto b}$. Once all variables have been processed, the resulting formula either contains the empty clause $\square$, and we declare this run of PPSZ a failure; or it does not, in which case PPSZ has found a satisfying assignment. If PPSZ has success probability p then we can repeat it $1/p$ times, obtaining a constant success probability. As long as P runs in subexponential time, the overall running time of this Monte Carlo algorithm is dominated by $1/p$ (which will, most likely, be exponential in n). Which proof

heuristics P should one consider? There are currently just two on the market. The first one is P_w, which checks whether $(x = b)$ is implied by a set of up to w clauses of Φ. The second one is R_w, which tries to derive the clause $(x = b)$ by resolution, bounded by width w. Obviously they both can be implemented in time $O^*(\binom{|\Phi|}{w}) \leq O^*(\binom{m^k}{w})$, which is subexponential as long as $w \in o(m/\log m)$. It is easy to see that $R_{w \cdot k}$ is at least as strong as P_w. We also speak of weak PPSZ when it uses P_w and strong PPSZ when it uses R_w (ignoring the concrete values of w).

2.1 Notation

Given a set of variables X, a partial assignment is a function $\alpha : X \to \{0, 1, *\}$, that is an assignment of 0-1 values to some of the variables with $*$ intended to mean unset by α. We denote the set of variables to which α gives a value by $\operatorname{var}(\alpha) := \{x \in X : \alpha(x) \in \{0, 1\}\}$. For two partial assignments α and β we write $\alpha \subseteq \beta$ to mean that for every $x \in \operatorname{var}(\alpha)$, it holds that $\beta(x) = \alpha(x)$. Naturally, $\alpha \subset \beta$ means that $\alpha \subseteq \beta$ and $|\operatorname{var}(\alpha)| < |\operatorname{var}(\beta)|$. Given a variable x and $b \in \{0, 1\}$, $x \mapsto b$ is the partial assignment which sets x to b. The assignment which sets every variable to0 is denoted by $\mathbf{0}$. For $Y \subseteq X$, we write $Y \mapsto 0$ to denote the partial assignment which sets all variables in Y to0. If $\operatorname{var}(\alpha) \cap \operatorname{var}(\beta) = \varnothing$, we define $\alpha \cup \beta$ to be the partial assignment which sets all $x \in \operatorname{var}(\alpha)$ to $\alpha(x)$, all $x \in \operatorname{var}(\beta)$ to $\beta(x)$, and all other variables to $*$. Finally the restriction of a formula Φ by α is denoted by $\Phi|_\alpha$.

2 Our Results

Theorem 2 (SSETH Holds for PPSZ). For every $k \in \mathbb{N}$, there is a polynomial p and a sequence $(F_m)_{m \in \mathbb{N}}$ of satisfiable k-CNF formulas F_m on m variables, such that for every $\epsilon > 0$ and

$$w \leq \sqrt{\epsilon \cdot \frac{\log \log m}{2 \log k}} - 3,$$

it holds that

$$\Pr[\text{ppsz}(F_m, R_w) \text{ succeeds}] \leq p(m) 2^{-m(1-2(1+\epsilon)/k)}.$$

Thus, the super strong exponential time hypothesis is true for Strong PPSZ, provided that we do not make it too strong, i.e., keep w fairly small. Note that this gives an upper bound on the savings of PPSZ by $2/k$, which is quite close to the currently best lower bound of $(\pi^2/6 + o(1))/k$ [7].

2.2 The Formula

Let $G = (V, E)$ be a graph. For every $e \in E(G)$ we introduce a variable x_e. Given a charge $c : V \to \{0, 1\}$, the Tseitin formula on G with charge c is the

Boolean formula

$$\text{Tseitin}(G,c) := \bigwedge_{u\in V(G)} \left(\sum_{e\in E(G):u\in e} x_e \equiv c(u) \pmod 2 \right). \tag{1}$$

If G has maximum degree k then this can be expressed as a k-CNF formula on $m = |E(G)|$ variables and $|V(G)|2^{k-1}$ clauses. Usually in proof complexity, the charge c is chosen so that $\text{Tseitin}(G,c)$ is unsatisfiable. In this paper, all charges will be0, and $\text{Tseitin}(G,0)$ is obviously satisfiable: set all variables to0. We will hence drop c from the notation and simply write $\text{Tseitin}(G)$ to denote this formula. The constraint $\sum_{e\in E(G):u\in e} x_e \equiv 0 \pmod 2$ is called the Tseitin constraint of vertex u. Given a set B of pairs of edges in G consider the following formula

$$\text{Tseitin}(G) \wedge \bigwedge_{\{e,f\}\in B} (\bar{x}_e \vee \bar{x}_f).$$

The constraint $(\bar{x}_e \vee \bar{x}_f)$ is called a bridge constraint. It is easy to see that $\mathbf{0}$ is the unique satisfying assignment of this formula if and only if every cycle in G contains a bridge in B. We will consider a particular instantiation of bridges given by graph homomorphisms. A graph homomorphism from a graph G to a graph H is a function $\phi : V(G) \to V(H)$ such that $\{\phi(u),\phi(v)\} \in E(H)$ whenever $\{u,v\} \in E(G)$. Thus, ϕ also induces a function from $E(G)$ to $E(H)$; $\phi(\{u,v\}) := \{\phi(u),\phi(v)\}$. Given G, H, and a homomorphism ϕ from G to H, we define a Tseitin formula with bridges on the variable set $\{x_e \mid e \in E(G)\}$:

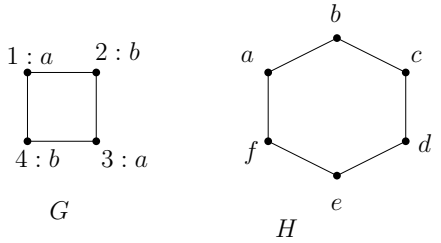
$$\text{TseitinBridge}(G,H,\phi) := \text{Tseitin}(G) \wedge \bigwedge_{\substack{e,f\in E(G)\\ e\neq f,\phi(e)=\phi(f)}} (\bar{x}_e \vee \bar{x}_f). \tag{2}$$

For brevity, we write $V = V(G)$ and $E = E(G)$.

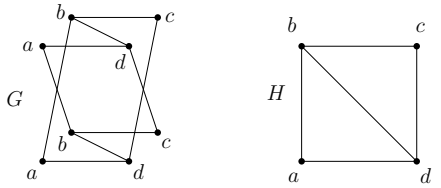
Observation 5. If $\text{girth}(G) > |E(H)|$ then $\text{TseitinBridge}(G,H,\phi)$ is uniquely satisfiable by $\mathbf{0}$.

Proof. Let $\alpha \neq \mathbf{0}$ be a total assignment. Let $F := \{e \in E(G) \mid \alpha(x_e) = 1\}$. If some vertex u has degree1 in (V,F), then α violates its Tseitin constraint. Otherwise, (V,F) has a cycle, which has length at least $\text{girth}(G)$. By the pigeonhole principle, this cycle contains two edges e, f such that $\phi(e) = \phi(f)$, and thus α violates their bridge constraint. □

Locally Injective Homomorphisms A homomorphism ϕ is called locally injective if for every $u \in V(G)$ and any two of its neighbors v_1 and v_2, it holds that $\phi(v_1) \neq \phi(v_2)$. Note that $\phi : G \to H$ being locally injective immediately implies that $\deg_G(u) \leq \deg_H(\phi(u))$. We call ϕ locally bijective if, additionally, $\deg_G(u) = \deg_H(\phi(u))$ for all vertices u of G. Note that a locally bijective homomorphism bijectively maps the neighborhood of u to the neighborhood of $\phi(u)$. The graph G is called a covering graph of H or a lift of H.



Example of a homomorphism that is not locally injective. The two neighbors of 1 are both mapped to b.



Example of a locally bijective homomorphism. The letters next to the vertices of G are not their names but rather their images under φ.

Theorem 6. Let G be a graph on n vertices and m edges. Suppose there is a locally injective graph homomorphism $\phi : G \to H$ for some graph H with $|E(H)| < \text{girth}(G)$. Then for all $\epsilon > 0$ and $w := \sqrt{\epsilon \cdot \text{girth}(H)/2} - 3$, the success probability of PPSZ with heuristic R_w on $\Phi := \text{TseitinBridge}(G, H, \phi)$ is at most

$$\Pr[\text{ppsz}(\Phi, R_w)] \leq 2^{-m+(1+\epsilon)n}.$$

Proof of Theorem2 using Theorem6. We first show how to construct F_m for infinitely many m. Let n_0 be some given, sufficiently large even integer. A well-known fact, first proven by Erdős and Sachs [4], is that there is a k-regular graph G_0 on n_0 vertices having girth at least $g_0 := \log n_0 / \log(k-1)$. Set $n_1 := \lfloor 2(g_0 - 1)/k \rfloor$ or $n_1 := \lfloor 2(g_0 - 1)/k \rfloor - 1$, whichever is even, and let G_1 be a k-regular graph on n_1 vertices, such that $\text{girth}(G_1) \geq g_1 := \log n_1 / \log(k-1)$. This exists, provided that n_0 is sufficiently large. Note that G_1 has at most $g_0 - 1 < \text{girth}(G_0)$ edges. A result by Angluin and Gardiner [1] states that there is a common lift G of G_0 and G_1. That is, G is a covering graph of G_0 and of G_1. Being a lift of a k-regular graph, G is k-regular as well. A closer inspection of their proof reveals that $n := |V(G)| \leq 4 n_0 n_1$. Let $m := kn/2$ be the number of edges in G. We set $\Phi_m := \text{TseitinBridge}(G, G_1, \phi_1)$, where ϕ_1 is the locally bijective homomorphism from G to G_1. It is not difficult to see that lifting cannot decrease the girth, and thus $\text{girth}(G) \geq \text{girth}(G_0) > |E(G_1)|$. Thus, we can apply Theorem6 to G, G_1, and ϕ_1, and conclude that the success probability of PPSZ on Φ_m is at most $2^{-m+(1+\epsilon)n}$ when using heuristic R_w. A quick calculation shows that $g_1 \geq \log\log m / \log k$ if n_0 is sufficiently large, and thus

$$w \geq \sqrt{\epsilon \cdot \frac{\log\log m}{2 \log k}} - 3.$$

This construction gives us an infinite set $M \subseteq \mathbb{N}$ and, for each $m \in M$, a satisfiable k-CNF formula F_m on m variables for which the claimed hardness result holds. By a simple tweaking of the construction, we can ensure that M is

"reasonably dense", meaning that there is some $m^* \in M \cap [m - \log m, m]$ for all sufficiently large m. We then let F_{m^*} be F_m, plus $m - m^*$ dummy variables. The success probability is then at most $2^{-m^*(1-2(1+\epsilon)/k)} \leq \text{poly}(m)2^{-m(1-2(1+\epsilon)/k)}$. We leave the details to the reader. □

3 All You Need to Know About PPSZ: Proof of Theorem6

We will explain the connection between PPSZ and width-bounded resolution lower bounds. After this section, the reader can forget everything about PPSZ and think of this paper as proving a certain resolution width lower bound. If $C = (C' \vee x)$ and $D = (D' \vee \bar{x})$ are clauses, then $(C' \vee D')$ is called the resolvent of C and D. It is clear that $C \wedge D$ logically implies $C' \vee D'$. Let Φ be a CNF formula. A resolution derivation from Φ is a sequence of clauses $C_1, \ldots, C_t$ such that every C_i is (1) a clause of Φ or (2) the resolvent of two earlier clauses. The width of the derivation is $\max_{1 \leq i \leq t} |C_i|$. For a clause C, we denote by $\text{width}(\Phi \vdash C)$ the minimum width of a resolution derivation from Φ that contains C. Resolution is complete for refutations, that is, Φ is unsatisfiable if and only if there is a derivation of the empty clause, denoted by $\square$, from Φ.

Proof of Theorem6. Let G, H, ϕ be as in Theorem6, and let $\Phi := \text{TseitinBridge}(G, H, \phi)$. The only satisfying assignment of Φ is $\mathbf{0}$. Consider a variant of PPSZ run on Φ such that whenever it has to pick a random value for a variable, it correctly sets it to0. Fix a permutation π. Let $Y(\pi)$ be the set of variables for which this variant of PPSZ under π could not derive the value using R_w, and let $Z(\pi) := \text{var}(\Phi) \setminus Y(\pi)$ be the rest, i.e., all variables whose value can be derived using R_w once all variables before them in π are set to0. It is not difficult to see that the success probability of the actual PPSZ on Φ is exactly $\mathbb{E}_\pi[2^{-|Y(\pi)|}]$. Suppose, for the sake of contradiction, that PPSZ using heuristic R_w has success probability greater than $2^{-m+(1+\epsilon)n}$. Then there is some π for which $Z(\pi) \geq (1+\epsilon)n$. Fix this π and set $Z := Z(\pi)$ and $Y := Y(\pi)$. The set of variables Z corresponds to a set F of edges, $F = \{e \in E(G) : x_e \in Z\}$. Set $G' = (V, F)$. Note that G' has n vertices and at least $(1+\epsilon)n$ edges. Setting a variable in Φ to0 corresponds to simply deleting the corresponding edge in G, and therefore

$$\Phi|_{Y \mapsto 0} = \text{TseitinBridge}(G', H, \phi).$$

For a graph $G = (V, E)$ and a set $X \subseteq V$, define the edge boundary $\partial(X) := \{e \in E : |e \cap X| = 1\}$. Call G an (a, b)-expander if $|\partial(X)| \geq b$ for all sets X of exactly a vertices. The next lemma is basically Lemma17 from [9], adapted for our purposes. We give a proof for completeness.

Lemma7. Let $\epsilon > 0$ and let G' be a graph on n vertices with at least $(1+\epsilon)n$ edges. Let $\ell \in \mathbb{N}$ and $h = \ell/\epsilon$. If $h < \text{girth}(G')$ then G' contains a non-empty subgraph G'' that has minimum degree at least2 and is an $(h, \ell+1)$-expander.

Proof. Start with $G'' = G'$. If G'' has a vertex of degree0 or1, delete it. If G'' contains a set X of h vertices with $|\partial(X)| \leq \ell$, delete X from G'', along with all incident edges. The first type of deletion removes one vertex and at most one edge. The second type removes exactly h vertices. There are at most

ℓ edges in the boundary of X; since $|X| < \text{girth}(G')$, the graph $G''[X]$ is a forest, and thus there are at most $h-1$ edges within X. Thus, removing X removes at most $\ell + h - 1 < (1+\epsilon)h$ edges. We see that a step that removes a vertices removes fewer than $(1+\epsilon)a$ edges. Suppose the process terminates with t vertices deleted. Trivially $t \leq n$. Fewer than $(1+\epsilon)n$ edges have been deleted, so G'' is non-empty. □

Let G'' be given by Lemma7 with $\ell := w+1$. We will further restrict Φ so that only edges of G'' remain unset. Let $F'' := E(G) \setminus E(G'')$, $Y'' := \{x_e : e \in F''\}$, and $\Phi'' := \Phi|_{Y'' \mapsto 0}$. Note that $\Phi'' = \text{TseitinBridge}(G'', H, \phi)$. Recall that all edges of G'' are mentioned in Z and since $Y'' \supseteq Y$ and restricting additional variables cannot increase the resolution width, we conclude that there exists $e \in E(G'')$ such that

$$\text{width}(\Phi'' \vdash \bar{x}_e) \leq w. \tag{3}$$

Towards a contradiction, we claim that in fact this resolution width is large for all variables x_e where $e \in E(G'')$. Indeed, we have the following theorem:

Theorem8 (Resolution Lower Bound). Let G be a graph of minimum degree2 that is an $(h, \ell+1)$-expander. Suppose there is a locally injective homomorphism $\phi : G \to H$ into some graph H. Then

$$\text{width}(\text{TseitinBridge}(G, H, \phi) \vdash \bar{x}_e) > \ell - 1, \tag{4}$$

for all edges e of G, provided that $2h\ell + 5h + \ell < \text{girth}(H)$.

Note that G'' has minimum degree2 and is a $(h, \ell+1)$-expander for $\ell = w+1$ and $h = (w+1)/\epsilon$. Also note that $\phi : V(G'') \to V(H)$ (or rather, the restriction of ϕ to $V(G'')$) is still a locally injective homomorphism. Recall that $w = \sqrt{\epsilon \cdot \text{girth}(H)/2} - 3$ and hence $2h\ell + 5h + \ell = 2(w+1)^2/\epsilon + 5(w+1)/\epsilon + w + 1 < \text{girth}(H)$, and thus Theorem8 applies to G''. This contradicts(3) and finishes the proof of Theorem6. □

4 Proof of Theorem8

Let $\Phi = \text{TseitinBridge}(G, H, \phi)$ and let e^* be an edge of G. We will show that $\text{width}(\Phi \vdash \bar{x}_{e^*}) > \ell - 1$ for all such edges e^*. In fact, we will prove $\text{width}(\Phi|_{x_{e^*} \mapsto 1} \vdash \square) > \ell - 1$, which is a slightly stronger statement. We will use the game characterization of resolution width due to Atserias and Dalmau [2]. Given a CNF formula F, the ℓ-bounded Atserias-Dalmau game played by two players, Prover and Delayer is defined as follows. A position in this game is a partial assignment α setting up to ℓ variables. The start position is the empty assignment. At position α, Prover can either (1) forget some variables, i.e., replace α by some $\beta \subset \alpha$. Or, (2), if $|\,\text{var}(\alpha)| \leq \ell - 1$, pick a variable $x \notin \text{var}(\alpha)$ and query it; Delayer has to respond with a truth value $b \in \{0, 1\}$, and α is updated to $\alpha \cup (x \mapsto b)$. The game ends if α violates a clause of F, in which case Prover wins. Delayer wins if she has a strategy to play indefinitely.

Theorem9 (Atserias and Dalmau [2]). Let F be an unsatisfiable CNF formula. If Delayer has a winning strategy for the $\ell+1$-bounded game then there is no width-ℓ resolution refutation of F.

To show that $\text{width}(\Phi|_{x_{e^*}\mapsto 1} \vdash \square) > \ell - 1$ we define a winning strategy for Delayer for the ℓ-bounded game that ensures she never loses. Indeed, we will modify the game a bit: it is now played on Φ instead of $\Phi|_{x_{e^*}\mapsto 1}$; the starting position is the partial assignment $x_{e^*} \mapsto 1$; Prover can never forget x_{e^*} but is now allowed partial assignments up to size $\ell + 1$. That is, he can query a new variable provided $|\operatorname{var}(\alpha)| \leq \ell$. It is easy to see that if Delayer wins this modified game, she wins the original one, too. Since $\Phi = \text{TseitinBridge}(G, H, \phi)$, we can easily rephrase the rules of the game in terms of sets of edges instead of partial assignments:

> **The Atserias-Dalmau, Graph View.** A position of the game is described by two disjoint set $F_0, F_1 \subseteq E(G)$. F_0 and F_1 correspond to the variables of Φ that the current partial assignment sets to0 and1, respectively. The start position is $F_0 = \varnothing$ and $F_1 = \{e^*\}$. In every step, Prover either (1) removes one edge e from F_0 or F_1 (but never removes e^*). Or (2) he queries an edge $e \in E(G) \setminus (F_0 \cup F_1)$, provided $|F_0| + |F_1| \leq \ell$. Delayer can then decide whether to add e to F_0 or F_1. Prover wins if there is a vertex u in G such that all edges incident to u are in $F_0 \cup F_1$ but $\deg_{F_1}(u)$ is odd (then the partial assignment α violates the Tseitin constraint of u); or if there are two edges $e, f \in F_1$ with $\phi(e) = \phi(f)$ (then α violates a bridge constraint).

We will now describe a winning strategy for Delayer. Throughout the game, she maintains a set $\widetilde{F}_1$ such that $F_1 \subseteq \widetilde{F}_1 \subseteq E \setminus F_0$. Let $V(\widetilde{F}_1)$ denote the set of vertices incident to at least one edge of $\widetilde{F}_1$. She makes sure $\widetilde{F}_1$ satisfies certain invariants:

1. Every connected component of $(V, \widetilde{F}_1)$ is a path; a path of positive length (i.e., a path that is not an isolated vertex) is called an $\widetilde{F}_1$-path.
2. Every $\widetilde{F}_1$-path contains at least one edge of F_1.
3. ϕ is injective on $V(\widetilde{F}_1)$.
4. Each $\widetilde{F}_1$-path has length at least $2h + 1$, and the first and last h edges of every $\widetilde{F}_1$-path are not in F_1.
5. Each $\widetilde{F}_1$-path has length at most $2h\ell + 2h + \ell$.

Observation10. If $\widetilde{F}_1$ satisfies the invariants, then no constraint is violated.

Proof. In fact we show that invariants1–4 already give the result. First, consider a Tseitin constraint of a vertex u. Since $\widetilde{F}_1$ consists of disjoint paths, so does F_1. Thus, u is incident to0,1, or2 edges of F_1. If it is incident to0 or2 edges of F_1, the Tseitin constraint of u is clearly not falsified. If it is incident to exactly one edge of F_1, then it is the endpoint of some path of F_1-edges. By Invariant4, u is incident to some other edge $f \in \widetilde{F}_1 \setminus F_1$. Thus, f is neither in F_0 nor in F_1, and the Tseitin constraint of u is not violated. Next, consider a bridge constraint $(\bar{x}_e \vee \bar{x}_f)$. By construction we have $\phi(e) = \phi(f)$. By Invariant3, ϕ is injective on $\widetilde{F}_1$, and thus e, f cannot both be in F_1, and the bridge constraint is not violated. □

We will use the following property of ϕ.

Proposition11. Let G' be a connected subgraph of G of diameter less than girth(H). Then ϕ is injective on $V(G')$, and thus $\phi(G')$ is isomorphic to G'.

Proof. For the sake of contradiction, suppose $u, v \in V(G')$ are two vertices with $\phi(u) = \phi(v)$. Let p be a shortest path from u to v in G'. Write p as $u = u_0, u_1, \ldots, u_t = v$. By assumption, $t <$ girth(H). Under ϕ, the path p is mapped to a reduced walk in H, reduced meaning that $\phi(u_{i-1}) \neq \phi(u_{i+1})$ for all $1 \leq i \leq t-1$. Since $\phi(u) = \phi(v)$, this is a closed walk and thus contains a cycle. The cycle has length at most $t <$ girth(H), a contradiction. □

How to initialize $\widetilde{F}_1$ Delayer can easily initialize $\widetilde{F}_1$. Write $e^* = \{u^*, v^*\}$. Since G has minimum degree2, Delayer can start a reduced walk from u^* of length h, and also from v^* and add this to $\widetilde{F}_1$. Since $2h + 1 <$ girth(H), this is a path; by Proposition11, ϕ is injective on its vertices.

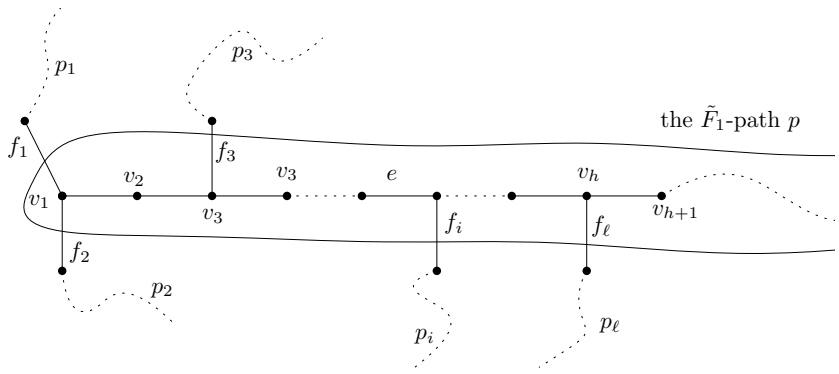
How to handle a Forget Step Suppose Prover forgets some edge $e \in F_0 \cup F_1$. If $e \in F_0$, Delayer leaves $\widetilde{F}_1$ unchanged. If $e \in F_1$, let p be the $\widetilde{F}_1$-path containing e. If p contains some other F_1-edge besides e, Delayer does not change $\widetilde{F}_1$; otherwise it simply removes all of p from $\widetilde{F}_1$. All invariants stay satisfied.

How to handle a Query from Prover Suppose Prover queries an edge e. Delayer has now to choose whether to include e into F_0 or F_1, and potentially update $\widetilde{F}_1$.

Case1: e is not in $\widetilde{F}_1$ Then Delayer adds e to F_0 and leaves $\widetilde{F}_1$ unchanged. All invariants still hold. This includes the case that e is incident to some vertex on a $\widetilde{F}_1$-path, but is not itself inside this path.

Case2: e is in some $\widetilde{F}_1$-path p but not among its first or last h edges Delayer adds e to F_1 and leaves $\widetilde{F}_1$ unchanged. All invariants still hold.

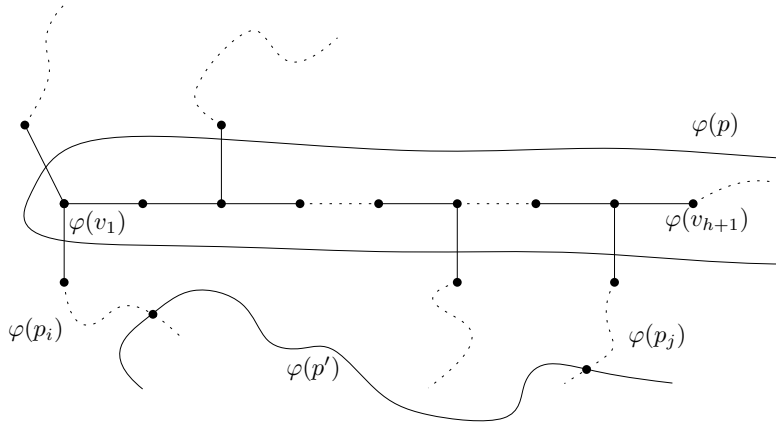
Case3: e is among the first or last h edges of some $\widetilde{F}_1$-path p Let $v_1, \ldots, v_{h+1}$ be the first $h+1$ vertices of p, and let q denote the length-h-path $v_1, \ldots, v_{h+1}$. By assumption, e lies on the path q. Since G is an $(h, \ell+1)$-expander, there are edges $f_1, \ldots, f_{\ell+1}$, each incident to exactly one vertex in $\{v_1, \ldots, v_h\}$. One of those edges could be $\{v_h, v_{h+1}\}$, but without loss of generality, for $1 \leq i \leq \ell$, edge f_i connects some $a_i \in \{v_1, \ldots, v_h\}$ to some b_i outside $\{v_1, \ldots, v_{h+1}\}$. Since G has minimum degree2 and girth larger than h, we can find paths $p_1, \ldots, p_\ell$ such that each p_i has length h and starts with a_i as its first and b_i as its second vertex. Since $3h <$ girth$(H) \leq$ girth(G), the p_i are vertex-disjoint. Since $h + |p| \leq h + 2h\ell + 2h + \ell <$ girth$(H) \leq$ girth(G), the path p_i intersects p only in vertex a_i. Thus, $C := p \cup p_1 \cup \cdots \cup p_\ell$ is a tree, and its diameter is at most $h + 2h\ell + 2h + \ell$. This figure shows how C could look like:



Call p_i blocked by F_0 if it contains some edge from F_0; at most $|F_0|$ of the ℓ paths are blocked by F_0. Let p' be an $\widetilde{F}_1$-path different from p. We say p' blocks p_i if the vertex sets of $\phi(p')$ and $\phi(p_i)$ intersect.

Proposition12. Let path p' in $\widetilde{F}_1$ be different from p. Then p' blocks at most one of the paths $p_1, \ldots, p_\ell$.

Proof. Let $C = p \cup p_1 \cup \cdots \cup p_\ell$. As argued above, this is a tree in G and its diameter is less than girth(H). By Proposition11, its image $\phi(C)$ is a tree in H, isomorphic to C. Suppose, for the sake of contradiction, that $\phi(p')$ intersects $\phi(p_i)$ and $\phi(p_j)$. This scenario would look like this:



Since $\phi(p')$ and $\phi(p)$ do not share any vertex (by Invariant3), the subgraph $\phi(p') \cup \phi(p_i) \cup \phi(p_j) \cup \phi(p)$ contains a cycle. This cycle has size at most $|p'| + |p_i| + |p_j| + |q| \leq 2h\ell + 2h + \ell + 3h$, a contradiction. □

Call p_i blocked by $\widetilde{F}_1$ if there is some $\widetilde{F}_1$-path different from p that blocks p_i. By Proposition12, at most $|F_1| - 1$ paths p_i are blocked by $\widetilde{F}_1$. Thus, a total of at most $|F_0| + |F_1| - 1 \leq \ell - 1$ of the paths p_i are blocked by F_0 or $\widetilde{F}_1$. Thus, there exists some path p_i, $1 \leq i \leq \ell$, that is not blocked. We now modify p by removing the edges on the path $v_1, v_2, \ldots, a_i$ and adding p_i. Let $\hat{p}$ denote the new version of p and $\hat{F}_1$ the new version of $\widetilde{F}_1$. Note that $F_1 \subseteq \hat{F}_1$ still holds, since we only modify the set $\widetilde{F}_1 \setminus F_1$. Obviously, $\hat{F}$ satisfies Invariants1,2, and4. Since p_i is not blocked by F_0, $\hat{F}$ is disjoint from F_0; because p_i is not

blocked by $\widetilde{F}_1$, Invariant3 still holds. Invariant5 might be violated: $\hat{p}$ might be too long. We will deal with this in a minute. Note that e is now either outside $\hat{F}_1$, and Delayer can include it into F_0; or it is inside $\hat{p}$, but then it is not among the first or last h edges of $\hat{p}$, and Delayer can include it into F_1. It remains to address the possibility that $\hat{p}$ is too long, violating Invariant5. If indeed $\hat{p}$ has more than $2h\ell + 2h + \ell$ edges, then it must somewhere contain $2h+1$ consecutive edges that are not in F_1 (note that $|F_1| \leq \ell$). Let $e_0, \ldots, e_{2h}$ be these edges. Define $\hat{F}_1 := \widetilde{F}_1 \setminus \{e_h\}$. That is, we split $\hat{p}$ into two parts, the first ending in $e_0, \ldots, e_{h-1}$, the second starting with $e_{h+1}, \ldots, e_{2h}$. Note that this satisfies Invariant4. If one of these paths contains no edge from F_1 at all, we delete it from $\hat{F}$. We continue this process until all paths in $\hat{F}$ have size at most $2h\ell + 2h + \ell$. The final $\hat{F}$ satisfies all invariants.

A Existence of Common Lift

Theorem13 (Angluin and Gardiner [1]). Let G and H be k-regular graphs. Then there exists a k-regular graph L that is a common lift of both G and H. Furthermore, $|V(L)| \leq 4|V(G)| \cdot |V(H)|$.

Proof. Suppose first that both $G = (U, E)$ and $H = (V, F)$ are bipartite. By Hall's Theorem, each has a perfect matching, and in fact, we can partition E and F into k perfect matchings each: $E = E_1 \uplus \cdots \uplus E_k$ and $F = F_1 \uplus \cdots \uplus F_k$. The common lift L has vertex set $U \times V$ and edge set

$$\bigcup_{i=1}^{k} \left\{ \{(u,v),(u',v')\} \in \binom{U \times V}{2} \mid \{u,u'\} \in E_i, \{v,v'\} \in F_i \right\}.$$

It is not difficult to see that the projections $\phi_G : (u,v) \mapsto u$ and $\phi_H : (u,v) \mapsto v$ are locally bijective homomorphisms from L into G and H, respectively. If G (or H or both) fails to be bipartite, we first replace it by its 2-lift G_2. The vertex set of G_2 is $U \times \{0,1\}$, and we form its edge set by creating, for each $\{u,v\} \in E$, two edges $\{(u,0),(v,1)\}$ and $\{(u,1),(v,0)\}$. The graph G_2 is bipartite, and projection to the first coordinate is a locally bijective homomorphism. Finally, observe that the composition of locally bijective homomorphisms is again a locally bijective homomorphism. Altogether, we can replace G and H by their respective 2-lifts G_2 and H_2; these are bipartite graphs, so we find a common lift L on $4|U| \cdot |V|$ vertices. □

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